

“Do You Need the Sage’s Tea or the Friend’s Cola” Exploring the Differential Healing Effects of Generative AI Conversational Styles

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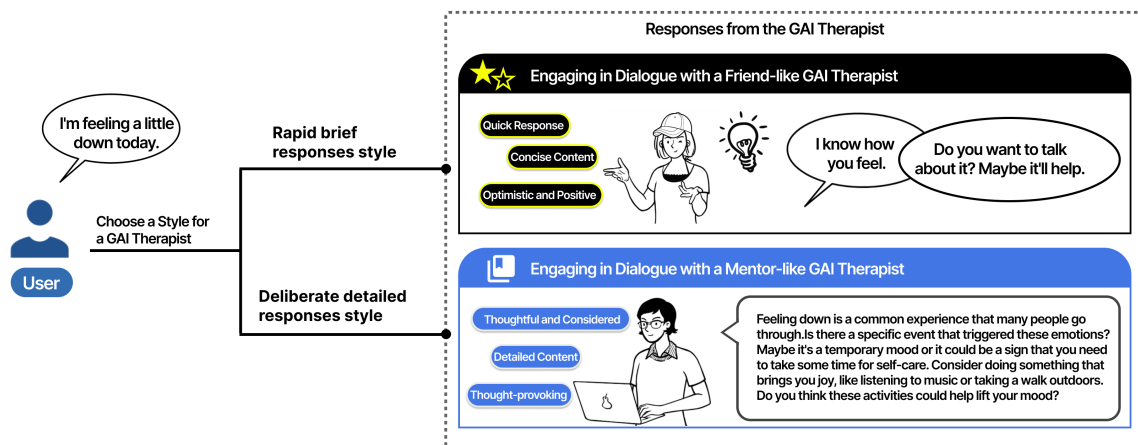


Figure 1: Two GAI therapist models with distinct styles

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Abstract

While the preliminary exploration of Generative AI in aiding psychotherapy has already commenced, there has been no research on the intrinsic selection mechanism on the user side during the process of GAI-assisted psychological healing. To address this gap, we experimented with 30 participants using the fine-tuned large language model, assessing emotional responses through heart rate variability, facial expressions, and SAM scales. Results show that participants engaging with GAI therapists exhibiting slow response times and rich content experienced higher happiness, while those interacting with faster, brief-content GAI therapists showed greater surprise. Additionally, despite the stress levels initially slightly rising during conversations with GAI therapists, they decrease over time and ultimately lead to an improved emotional state. These findings validate the effectiveness of GAI therapists and provide a basis for applying GAI in different psychotherapy scenarios.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; User studies.

Keywords

Human-AI interaction, Affective healing, Dialogue Styles, Facial expression, Heart rate variability

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1 INTRODUCTION

In the current digital era, Generative AI (GAI) has gradually integrated into various aspects of people’s lives, particularly demonstrating significant potential in the field of mental health [1–3]. With the acceleration of life pace, individuals are facing increasing levels of stress, leading to more prevalent mental health issues, while traditional mental health services are limited in resources [4]. Additionally, concerns about the privacy of such services often cause apprehension among individuals [5]. Against this backdrop, GAI offers new avenues for mental health therapy through its convenience, immediacy, and privacy.

Currently, the application of GAI in the field of mental health predominantly focuses on providing conversational support. However, the specific factors within these conversations that influence the healing process and the ultimate therapeutic outcomes remain largely underexplored. While user testing results of GAI-based therapeutic robots have demonstrated the effectiveness of this approach [6, 7], these studies predominantly rely on self-reports, which are vulnerable to influences from memory, emotions, and social desirability, making it difficult to accurately reflect the true therapeutic effects [8].

In response, this paper proposes an investigation into the impact of conversational style of GAI on therapeutic outcomes based on

physiological data, as seen in Figure 1. The research questions are as follows:

RQ1: Does the positive impact of GAI on emotional healing increase continuously over the course of the interaction?

RQ2: What impact do different conversational styles of GAI therapists have on the therapeutic outcomes?

To investigate the aforementioned questions, we designed an experiment to collect and analyze physiological data from participants interacting with GAI therapists exhibiting different conversational styles. By integrating these data with psychological scale assessments, we precisely elucidated the intrinsic relationship between conversational style and therapeutic outcomes. Our contributions are as follows:

Validation of GAI therapeutic effectiveness. Through the integration of physiological data and subjective scales, our experimental approach demonstrates that individuals experience positive emotional changes after interacting with GAI-based healers.

Identification of temporal changes in stress during GAI therapy. During the GAI therapeutic process, participants exhibited an initial increase in stress levels followed by a subsequent decrease. This pattern suggests that individuals may initially experience fear and distrust when interacting with GAI therapists, but these feelings tend to dissipate over time, ultimately achieve healing effect.

Provision of application guidelines for GAI in diverse therapeutic scenarios. For therapeutic contexts requiring long-term stress relief and a deep, immersive healing atmosphere, a combination of slow response speed and detailed answers is recommended. Conversely, for scenarios necessitating rapid emotional activation and increased vitality, a combination of fast response speed and brief answers is more suitable.

2 RELATED WORK

2.1 Conversational GAI Agent as a Psychological Therapist

The application of conversational emotion-supportive GAI in psychotherapy is emerging, demonstrating advantages in privacy, sustainability, and personalization [9, 10]. Using techniques such as Natural Language Processing, deep learning, and affective computing, GAI can simulate human-like conversations and emotional responses [11–13]. Mental health chatbots such as Woebot and Wysa have been employed to assist users in managing anxiety, depression, and emotional fluctuations by guiding them through daily conversations to engage in emotional reflection and behavioral adjustments [14, 15]. These AI systems not only emulate certain functions of human therapists but also provide a confidential and low-pressure communication channel, effectively reducing the stigma associated with psychotherapy [16].

2.2 Designing the Conversational Styles of GAI for Psychotherapeutic Scenarios

Despite advancements in natural conversational abilities of chatbots through deep learning, their responses remain monotonous and repetitive, impeding user engagement and causing frustration [17]. This limitation has prompted the development of empathetic

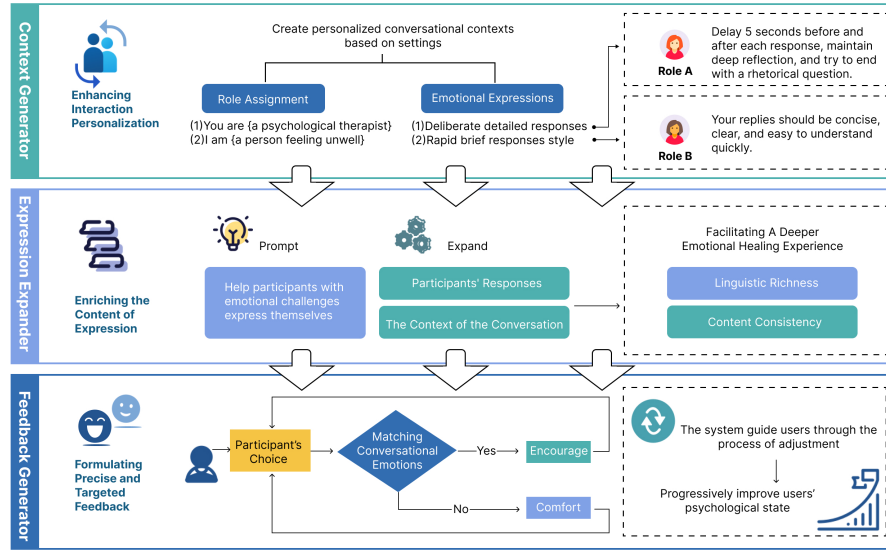


Figure 2: The three components that constitute the GAI therapist model

chatbots, which are designed to understand human emotions in context and generate contextually appropriate responses to provide genuine emotional support to users.

Recent studies have explored the application of GAI chatbots across diverse contexts. For example, an empathetic chatbot reduced negative emotions among socially isolated individuals [18]. A proactive smoking cessation chatbot, which provided direct and reality-oriented feedback, effectively facilitated user self-awareness [19]. Additionally, a multimodal chatbot designed for stroke patients offered personalized interaction options tailored to different stages of recovery [20].

2.3 Experimental Validation of GAI Chatbots in Mental Health

2.3.1 Physiological Measurements. Physiological measures, such as Heart Rate Variability (HRV), offer an objective assessment of the therapeutic impact of GAI. HRV, defined as the variation in time intervals between successive heartbeats, is a robust indicator of stress and autonomic nervous system balance [21–23]. In studies involving GAI-assisted therapy, wearable devices have been employed to monitor HRV during user interactions with AI chatbots. Results indicate that empathetic conversations with AI chatbots are associated with increased HRV, reflecting reduced stress levels [24].

2.3.2 Facial Expression Data. The emergence of advanced computer vision and machine learning algorithms has elevated facial expression data analysis to a crucial tool in studying GAI-assisted therapeutic interventions [25]. In human-AI interactions, facial expression analysis provides real-time feedback on users’ emotional responses to AI outputs, thereby identifying conversational strategies that elicit positive emotions [26]. As immediate physiological responses, facial expressions can reflect even subtle changes in psychological states [27–29]. The Facial Action Coding System (FACS)

enables the classification of human facial muscle movements and is used to identify and interpret facial action units that underlie complex expressions [30, 31].

3 METHOD

3.1 Design of GAI Therapist

In this study, we designed a fine-tuned large language model capable of fulfilling the conversational tasks as required. As illustrated in the figure, researchers configured the chatbot settings based on the fine-tuning materials. Three key technical components were employed: the Context Generator, the Expression Expander, and the Feedback Generator [32]. These components work in concert to provide personalized emotional support and psychological counseling, as seen in Figure 2.

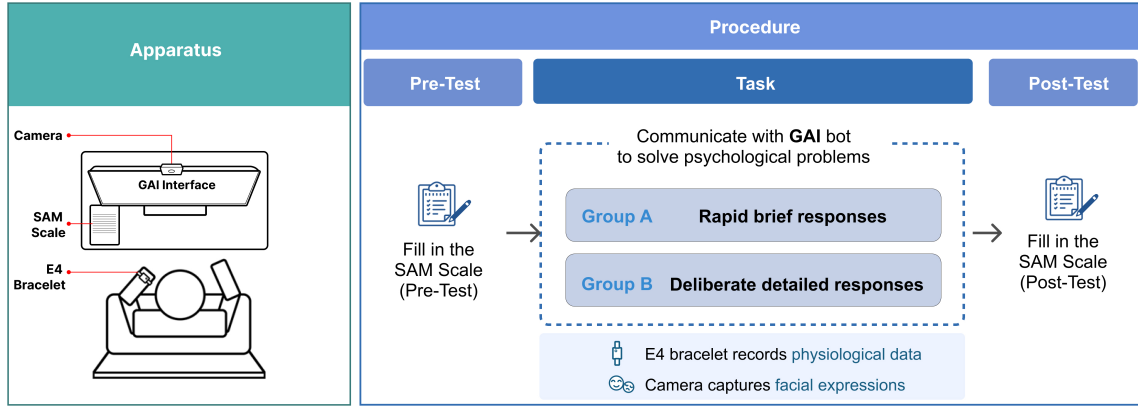
The model comprises two distinct styles of therapeutic consultants. Table 1 provides a summary of the configurations of the two GAI therapists. After the GAI-based therapeutic model was trained, participants were invited to interact with it, using voice-based conversations to assist users in addressing their psychological problems.

3.2 Participants

Before the experiment, we distributed the GAD-7 questionnaire, which is used for screening mental disorders in primary care settings [33]. Thirty participants (aged 18–30, $M = 22.97$, $SD = 2.81$) whose scores ranged from 5 to 14 (indicating mild to moderate anxiety or depression) were included in the study. The demographic data of the participants were tested, displaying no significant clustering characteristics. All procedures and experimental protocols were adhered to the ethical standards outlined in the 1975 Declaration of Helsinki.

Table 1: Model parameters

Category	Rapid brief responses style	Deliberate detailed responses style
Model	MiniMax 6.5s(245K)	Doubao-Function call model(32K)
Feature	Quick response	Strong ability to analyze complex tasks
Generation randomness	0.5	0.29
Top P cumulative probability	1	0.7
Number of context rounds	20	20
Maximum reply length	50	200

**Figure 3: Apparatus and Procedure of the experiment****Table 2: Data measurement indicators**

Metrics	Measurement	Sampling rate	Description
Discrete facial emotion data	Py-Feat Toolbox	64Hz	Mapping specific emotional categories using AUs activation patterns [34]
Heart rate variability (HRV)	E4 wristband	4Hz	Used to assess users' psychological stress levels [35, 36]
Valence; Arousal; Dominance	SAM Scale	/	Calculate user emotional state through 9-point Likert scale

3.3 Apparatus and procedure

Participants were seated comfortably in a natural posture to complete the experiment. A display screen was placed on the table in front of the participants, with a camera mounted at the top edge of the computer screen, adjusted to an appropriate angle to ensure comprehensive recording of the participants' facial information. The camera captured the participants' facial expressions by the FACS toolbox developed based on OpenCV [34]. Additionally, participants' heart rate variability data were recorded using an E4 wristband. The experimental setup and procedure were illustrated in Figure 3.

Participants initially completed the Pre-Test SAM scale and had their facial baseline expressions recorded. They then engaged in a dialogue with the GAI therapist assigned to their experimental group to address the issues at hand. During the interaction, facial

expression data and Heart Rate Variability (HRV) data were continuously recorded. After the dialogue, participants completed the Post-Test SAM scale.

4 DATA COLLECTION AND ANALYSIS

The present study primarily employed quantitative methods, as illustrated in Table 2. Quantitative data encompassed facial expressions, HRV data, and SAM scale.

4.1 Emotional Changes — For RQ1

The monitoring of participants' emotional changes and the validation of GAI therapeutic effectiveness are primarily conducted through the following two types of data:

The SAM scale. The SAM scale [37] assesses participants’ psychological experience of a stimulus across three dimensions: valence, arousal, and dominance. This subjective scale allows participants to document their emotional changes before and after exposure to the stimulus.

HRV physiological data. By obtaining interbeat interval data and importing it into Kubios HRV analysis software [38], the software can detect R-peaks and generate a time series of RR intervals. Here, the interbeat interval refers to the time span between two consecutive R-waves, also known as the RR interval. We used PNN50 as the key metric for assessing HRV, thereby evaluating stress levels. Under high psychological stress, the value of the PNN50 index decreases. Through this physiological data, we obtained the temporal changes in participants’ stress levels throughout the experiment.

4.2 Differences in the Healing Effects of GAI Dialogue Styles — For RQ2

Facial expression data. The FACS toolbox was used to collect facial expression data. It comprises six basic emotions derived from 20 fundamental action units [39]. By analyzing facial expression data following interactions with different conversational styles, we assessed the impact of conversational style on participants’ emotions.

5 RESULT

5.1 Emotional Changes — For RQ1

5.1.1 Self-Assessment Manikin Scale. As depicted in Figure 4, the paired results between Pre-Test and Post-Test for participants reveal the overall emotional changes before and after their interaction with the GAI therapist.

Valence: There was a significant difference ($p < 0.01$, Cohen’s $d = 0.913$) between the Pre-Test ($M = 4.30$, $SD = 2.07$) and the Post-Test ($M = 6.167$, $SD = 2.019$), indicating a significant increase in valence following the interaction.

Arousal: No significant difference was found ($p = 0.655$, Cohen’s $d = 0.039$) between the Pre-Test ($M = 4.167$, $SD = 0.874$) and the Post-Test ($M = 4.20$, $SD = 0.847$).

Dominance: A significant difference was observed ($p < 0.01$, Cohen’s $d = 4.296$) between the pre-test ($M = 3.633$, $SD = 0.615$) and the post-test ($M = 6.533$, $SD = 0.73$), indicating a significant increase in dominance following the interaction.

5.1.2 Physiological data: heart rate variability. The PNN50 index derived from HRV data further elucidates the temporal changes in stress levels among participants (see Figure 5). Each process was sequentially divided into eight segments, with the mean value of data from each segment used for analysis. The PNN50 values of each participant were compared with their respective baselines to demonstrate changes. Given that the results met the assumption of normality but not the assumption of homogeneity of variances, and considering the need to compare eight sampling points with the baseline, the Friedman test was employed. It revealed that during the conversation, the PNN50 value at the second sampling point significantly decreased compared to the baseline ($p < 0.01$), followed by a gradual increase, indicating an initial rise in stress that subsequently diminished throughout the process.

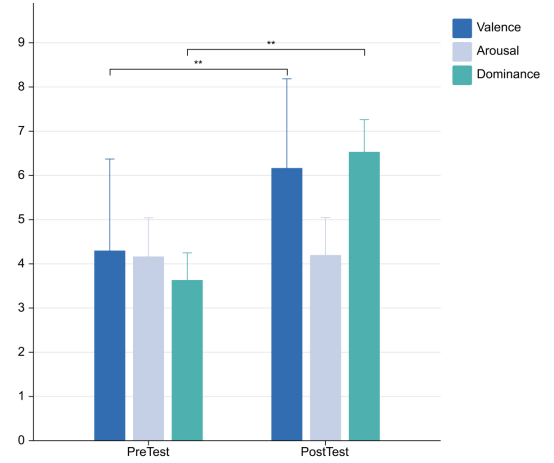


Figure 4: Differences between Pre-Test and Post-Test in SAM Scale

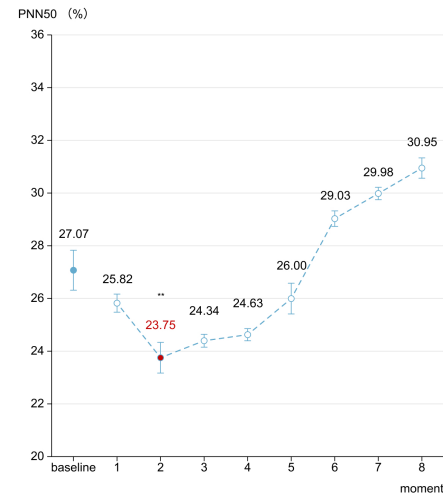


Figure 5: Changes in HRV among participants during communication with the GAI therapist

5.2 Differences in the Healing Effects of GAI Dialogue Styles — For RQ2

We tested two GAI therapists’ styles: “rapid and brief responses”, and “deliberate and detailed” responses. Before the experiment, we measured participants’ baseline emotions using a Friedman test (since data wasn’t normally distributed). Results showed no significant differences in facial expressions ($p = 0.553$), indicating similar initial emotions.

Given that the facial expression data did not conform to the normality criterion, a Mann-Whitney U test was performed for independent samples. The results revealed significant differences between the two response types in terms of surprise ($p < 0.01$, Cohen’s $f = 1.482$) and happiness ($p = 0.018$, Cohen’s $f = 1.317$).

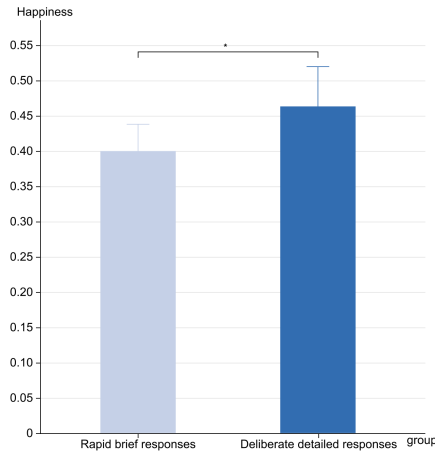


Figure 6: Happiness levels appear in different style of communication with GAI therapist

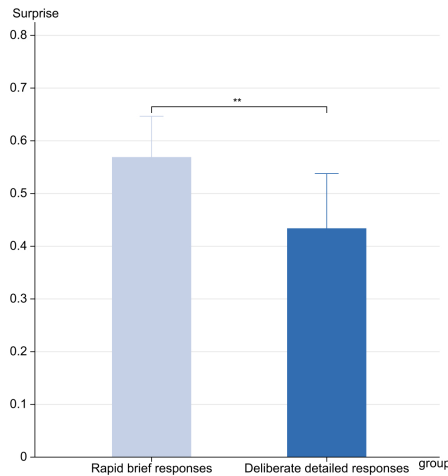


Figure 7: Surprise levels appear in different style of communication with GAI therapist

For happiness (Figure 6), the deliberate and detailed response group ($M = 0.37$, $SD = 0.045$) exhibited significantly higher levels of happiness compared to the rapid and brief response group ($M = 0.319$, $SD = 0.03$).

Conversely, for surprise (Figure 7), the rapid and brief response group ($M = 0.569$, $SD = 0.077$) demonstrated significantly higher levels of surprise compared to the deliberate and detailed response group ($M = 0.433$, $SD = 0.104$).

6 DISCUSSION

6.1 Verifying the Effectiveness of GAI Therapy from temporal changes

Following interactions with the GAI therapist, participants demonstrated significant improvements in emotional valence and dominance, aligning with recent findings in the field [40–42] and thereby

supporting our hypothesis regarding the efficacy of GAI-based interventions. Specifically, participants exhibited increased positive emotions and enhanced ability to manage the pace of communication with the GAI therapist.

Despite the overall positive outcomes, a detailed examination of the conversational process revealed that participants' stress levels did not consistently decrease throughout the interaction. Instead, stress levels, as indicated by HRV data, initially increased before gradually declining. This pattern suggests that initial discomfort and fear stemming from unfamiliarity with the GAI therapist contributed to elevated stress levels. However, as participants became more accustomed to the interaction, their stress levels were alleviated. Over time, participants grew more comfortable with the GAI therapist, ultimately leading to a reduction in negative emotions and stress levels compared to pre-communication baselines.

6.2 The Impact of Diverse Conversational Styles of GAI on Specific Healing Effects

Different communication styles employed by GAI in conversations demonstrate unique potential in emotional therapy. The study found that “rapid and brief responses” and “deliberate and detailed responses” have distinct emphases in regulating participants' emotions.

The “rapid and brief response” style of the GAI therapist, akin to a friend who responds quickly and provides commentary, readily elicits surprise in participants. From an information-processing perspective, frequent and brief feedback offers participants a continuous stream of novel stimuli, maintaining emotional tension and dynamic changes. This high-frequency feedback activates emotional response mechanisms, consistent with Lazarus's theory of emotion [43], which posits that frequent emotional stimuli can trigger more intense emotional reactions. The “rapid and brief response” style is particularly suitable for therapeutic scenarios requiring rapid emotional activation and increased vitality. By engaging in frequent interactions over a short period, it guides participants from a passive emotional state to a more active emotional experience.

In contrast, the “deliberate and detailed response” style of the GAI therapist, resembling a mentor who listens attentively and provides thoughtful answers, enhances participants' sense of well-being. This style offers participants more space for emotional processing and reflection, helping them to gradually relax at a steady pace. Self-Determination Theory suggests that individuals can better achieve emotional self-regulation and control in a safe and comfortable environment [44]. Slower responses increase participants' sense of trust, while detailed answers provide clearer guidance and ample time for understanding and reacting, thereby allowing for more composed emotional adjustment. This style is applicable to scenarios requiring long-term stress relief and a deep, immersive therapeutic atmosphere.

7 LIMITATIONS AND FUTURE WORK

Our study focused exclusively on two classic conversational styles: deliberate and detailed responses versus rapid and brief responses. Future research should explore a broader range of conversational approaches, including variations in tone, speech rate, and modes of

expression, to assess their adaptability across different therapeutic contexts.

The observed pattern of initial stress increase followed by a decrease during interactions with GAI therapists highlights several psychological questions regarding human-AI interactions. This finding resonates with recent studies, which suggest that although the use of GAI can activate physiological states, it may also induce psychological stress during the process [45]. Future work could investigate strategies to mitigate this initial stress spike, thereby reducing discomfort and accelerating the acceptance of GAI-based interventions.

Lastly, given the limited duration of our experimental sessions, the point at which stress levels stabilize after an initial increase remains vague. Further research is needed to determine the optimal timing and duration of GAI-based therapeutic sessions, which could inform the design of more effective interventions.

8 CONCLUSION

This study is dedicated to exploring user-side feedback on GAI-based conversational psychological healing. We began by designing two GAI therapists based on fine-tuning. Subsequently, we recruited 30 participants who interacted with both GAI therapists, during which we collected their SAM scale data, HRV data, and changes in facial expressions. Ultimately, through quantitative analysis, our study validated the efficacy of GAI-based healing, revealed the patterns of stress changes during the healing process, and provided a basis for selecting GAI healing methods across different contexts. Our future work will encompass a larger user population and investigate the adaptability of conversational approaches across diverse scenarios. Overall, our work offers a novel perspective and additional data support for the future application of GAI in the field of mental health therapy.

Acknowledgments

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